

ZHI ZHENG

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EDUCATION

National University of Singapore (NUS), Singapore

Aug. 2024 - Now

Ph.D. in Computer Science, NUS Research Scholarship.

Mentored by: Prof. Wee Sun Lee.

Southern University of Science and Technology (SUSTech), China

Sep. 2020 - Jun. 2024

B.Eng. in Intelligent Science and Technology, GPA: 3.62/4.00; Major GPA: 3.72/4.00.

Mentored by: Prof. Zhenkun Wang, Prof. Xin Yao, and Prof. Ke Tang.

RESEARCH INTERESTS

- **Large Language Model (LLM):**

LLM Reasoning, Latent Chain-of-Thought, Retrieval-augmented Generation.

- **Application of Deep Learning on Optimization:**

Neural Combinatorial Optimization, LLM for Optimization Tasks, Multi-objective optimization.

ACCEPTED PUBLICATIONS

- **Zhi Zheng**, Zhuoliang Xie, Zhenkun Wang, Bryan Hooi⁺; Monte Carlo Tree Search for Comprehensive Exploration in LLM-Based Automatic Heuristic Design; *ICML*, 2025.
- **Zhi Zheng**, Changliang Zhou, Xialiang Tong, Mingxuan Yuan, Zhenkun Wang⁺; UDC: A Unified Neural Divide-and-Conquer Framework for Large-Scale Combinatorial Optimization Problems; *NeurIPS*, 2024.
- **Zhi Zheng**, Shunyu Yao, Zhenkun Wang⁺, Xialiang Tong, Mingxuan Yuan, Ke Tang; DPN: Decoupling Partition and Navigation for Neural Solvers of Min-max Vehicle Routing Problems; *ICML*, 2024.
- Rui sun, **Zhi Zheng**^{*}, Zhenkun Wang⁺; Learning Encodings of Constructive Neural Combinatorial Optimization Needs to Regret; *In Proceedings of the AAAI Conference on Artificial Intelligence*, 2024.
- **Zhi Zheng**, Shunyu Yao, Genghui Li, Linxi Han, and Zhenkun Wang⁺; Pareto Improver: Learning Improvement Heuristics for Multi-Objective Route Planning; *IEEE Transactions on Intelligent Transportation Systems*, 2023;

* Equal contribution

⁺ Corresponding author

RESEARCH ACTIVITY

- **Reviewer:** IEEE trans journals, NeurIPS 2024, ICLR 2025, ICML 2025.

Aug. 2023 - present

HONORS & AWARDS

- Outstanding Graduation Thesis of Southern University of Science and Technology. Jun. 2024
- Outstanding Graduates of Southern University of Science and Technology. Jun. 2024
- **Second prize** in the ACM content of SUSTech. Jan. 2021
- **First prize** in the 23rd National Olympiad in Informatics in Provinces (NOIP2018). Oct. 2018

WORK EXPERIENCE

Shandong Inspur Scientific Research Institute, China

July 2022 - Aug. 2022

- Worked on channel pruning for deep learning computer vision models.

Teaching Assistant of CS3244 NUS, Singapore

Jan. 2025 - Apr. 2025

- Taught tutorial classes and scored projects.